OMAR HISHAM ZAKARIA MAHMOUD

GIS Specialist | GIS Analyst | Enterprise GIS | Utility Network | Geospatial Data Management

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Professional Summary

GIS Analyst with 6+ years of professional GIS experience in GIS data management, spatial analysis, mapping, and geospatial data quality across utility, infrastructure, urban planning, and enterprise GIS projects in Saudi Arabia and Egypt. Strong proficiency in the Esri GIS ecosystem, including ArcGIS Pro, ArcGIS Enterprise, ArcGIS Online, Enterprise Geodatabases, Utility Network, Data Reviewer, Attribute Rules, Topology, Model Builder, Experience Builder, and ArcGIS Dashboards. Experienced in spatial analysis, data validation, GIS QA/QC, geodatabase management, CAD-to-GIS conversion, as-built data integration, site suitability analysis, data editing, visualization, and GIS data publishing. Practical knowledge of FME, Python for GIS automation, SQL, QGIS, ENVI, AutoCAD, and 3D/mobile mapping. Skilled in transforming complex spatial data into accurate, structured, and actionable geospatial information to support infrastructure, utilities, asset management, urban planning, and operational decision-making.

Professional Experience

Riyadh Region General Secretariat Municipality and Privatization Agency

GIS Specialist — Jan 2026 – Present

- Managed and maintained GIS databases and enterprise geodatabases for land contracts, ATM and bank licenses, telecommunication tower contracts, utilities, and infrastructure assets, ensuring data accuracy, consistency, integrity, and completeness.
- Supported ArcGIS Platform projects using ArcGIS Pro, ArcGIS Enterprise, and ArcGIS Online for geospatial data management, editing, analysis, validation, and publishing.
- Performed spatial analysis and site suitability assessments to support infrastructure planning, operational requirements, and location-based decision-making.
- Supported utility network GIS workflows, including utility data management, network data validation, topology, attribute rules, and enterprise geodatabase workflows.
- Applied Data Reviewer, Topology, Attribute Rules, and Model Builder to perform GIS QA/QC, data validation, and quality control, improving data accuracy and workflow efficiency.
- Performed CAD-to-GIS and as-built-to-database data conversion and integration to support accurate representation of infrastructure and utility assets.
- Utilized Python and SQL for GIS data processing, workflow automation, validation, and quality control activities.

- Prepared, formatted, and delivered GIS datasets in Excel and KMZ formats, created professional map layouts, and integrated GIS outputs into tender plans and technical documentation.

- Prepared and published finalized GIS datasets through ArcGIS Enterprise and ArcGIS Online, ensuring accurate and accessible geospatial information for stakeholders and operational use.

- Prepared GIS maps, technical reports, presentations, and structured geospatial deliverables for responsible authorities, stakeholders, and project requirements.
- Collaborated with internal teams, consultants, contractors, and stakeholders to collect, validate, analyze, and deliver geospatial and utility data in accordance with project requirements and GIS standards.

General Organization for Physical Planning

GIS Expert Assistant Jan 2025 – Dec 2025.

- Created and maintained urban sector GIS layers from base maps and populated descriptive attribute tables to support urban planning and spatial data management.
- Prepared GIS datasets in Excel and KMZ formats, produced map layouts, and generated geospatial outputs for visualization and review in Google Earth.
- Supported spatial analysis, mapping, data preparation, and GIS documentation for urban and regional planning requirements.

The Modern Natural Gas Company

GIS Analyst – Quality Control Aug 2023 – Dec 2024

- Performed GIS Quality Control (QC) on point, line, and polygon datasets, ensuring accuracy, consistency, completeness, spatial data integrity, and compliance with project standards.
- Applied Esri Data Reviewer, Attribute Rules, and Topology Rules to validate spatial and attribute data, identify quality issues, and support corrective actions.
- Supported natural gas utility network GIS workflows, including CAD-to-GIS conversion, Utility Network, network topology, load analysis, test sections, fittings, network data entry, and Material Take-Off (MTO).
- Managed and validated utility GIS databases and enterprise geodatabases supporting natural gas network and infrastructure asset data.
- Used ArcGIS Pro and ArcGIS Enterprise for GIS data editing, validation, analysis, management, and publishing.
- Published GIS data through ArcGIS Enterprise Portal and supported visualization and analysis using Experience Builder and ArcGIS Dashboards.
- Developed geoprocessing workflows with ArcGIS ModelBuilder and utilized FME Data Tools and Python for spatial data transformation, workflow automation, processing, and GIS problem-solving.
- Performed CAD-to-GIS and As-Built-to-Database conversion for utility and infrastructure datasets.

- Contributed to GIS projects including Digital Transformation, Haya Karima, Jubail City, Crossing, and As-Built-to-Database conversion.

Micro Engineering Technic (METI)

GIS Specialist May 2023 – Aug 2023

Used Mobile Mapping Systems to label objects in 3D and verify the accuracy and quality of data and maps, ensuring their consistency with established standards to detect and correct errors, and improve the data collection and management process by using different software and online tools to work on Smart City projects.

EKC Geographic Information Systems

GIS Specialist – Saudi electricity project Jun 2021 – Dec 2021

Drawing and entering data for LV/MV lines, meters, and transformers.

Project: Jubail City – Hafar Al-Batin – Qaryat Al Ulya

Farsi Geotech

GIS Specialist - Saudi electricity project Jan 2020 – May 2021

Expanding the electrical network in A Saudi project

EDUCATION

Helwan University. **2021**

Bachelor's Degree in Geography and Geographic Information Systems (GIS)
Maps and Survey Department, Faculty of Arts.
Cumulative Grade: Very Good.

Technical Skills.

- Esri: ArcGIS Pro | ArcGIS Enterprise | ArcGIS Online | ArcGIS Utility Network | ArcGIS Dashboards | Experience Builder | Data Reviewer | Attribute Rules | Model Builder
- Automation & Data: Python | SQL | FME Data Tools
- GIS & Engineering: QGIS | ENVI | AutoCAD | Google Earth | KMZ
- Productivity tools: Microsoft Excel | Microsoft Word | Microsoft PowerPoint

CERTIFICATION

- **SAUDI CONCIL OF ENGINEERS** – *SPECIALIST 2026*
- **GENERAL ORGANIZATION FOR PHYSICAL PLANNING** – *EXPERT ASSISTANT 2025*
- **AFRICAN CENTRE FOR SURVYING SERVICES** – *SUMMER INTERNSHIP 2018*
- **THE ARAB CONTRACTORS** – *SUMMER INTERNSHIP 2019*
- **ESRI** – *10 CERTIFICATED GIS COURSES*

- **GEOSTEPS ACADEMY** – *2 CERTIFIED GIS COURSES*
- **UDEMY** – *2 CERTIFIED GIS COURSES*
- **COURSERA** – *2 CERTIFIED GIS COURSES*
- **EDRAAK** – *CERTIFIED AI COURSE*

Core Competencies.

- GIS Analysis & Spatial Analysis
- GIS Data Management & Geodatabase Management
- Enterprise GIS & ArcGIS Platform
- Utility Network & Utility GIS
- GIS Quality Assurance & Quality Control (QA/QC)
- Spatial & Attribute Data Validation
- Spatial Data Integrity & Data Quality
- Geodatabase Design & Management
- CAD-to-GIS & As-Built-to-Database Conversion
- GIS Workflow Automation & Geoprocessing
- Site Suitability & Location Analysis
- GIS Data Editing & Publishing
- Network Topology & Attribute Rules
- GIS Data Transformation & ETL
- Cartography & Geospatial Visualization
- Urban GIS & Infrastructure Mapping
- Utility & Infrastructure Asset Data
- Technical GIS Reporting & Documentation